Work Integrated Learning Programmes Division
M.Tech (AIML/SEZG) — AIML ZG521: NLP Applications

Assignment–2 (Total: 15 Marks)

Part 1: Programming (10 Marks) | Part 2: Situated Learning (5 Marks)

PART 1 — PROGRAMMING ASSIGNMENT (10 MARKS)

Statistical Machine Translation — From Word Alignment to Phrase-Based Translation and Neural MT Comparison

1. Objective

This assignment gives students hands-on experience building a Statistical Machine Translation (SMT) pipeline from first principles and comparing it against modern neural approaches. Each group will build a domain-specific translation system through three progressively advanced stages — word-level alignment using IBM Model 1 (Expectation–Maximization), a phrase-based SMT system with an n-gram language model and decoder, and a comparative evaluation against a pretrained Neural MT (NMT) model — and evaluate them on translation quality (BLEU), adequacy, fluency, alignment quality, latency, and robustness, to determine the strengths and limitations of statistical versus neural approaches for their assigned business scenario.

2. Problem Statement

PS	Domain	Business Scenario	Language Pair & Suggested Parallel Corpus	Stress-Test Focus
PS 4	News / Media	A news agency wants to translate wire stories and headlines from English to German to syndicate content to European partners.	**English → German** Europarl, News Commentary / WMT news data (sampled subset), OPUS corpora, or instructor-approved synthetic parallel sentences	Headlines with dropped articles/verbs, long compound German words, named entities and quotations, date and place expressions

Parallel Corpus Requirement

Each group must prepare a parallel corpus of approximately 5,000–20,000 sentence pairs (sampled from public corpora or synthetic, wherever feasible), plus a held-out test set of at least 200 sentence pairs. Acceptable sources include: public parallel corpora; government/organizational bilingual documents; bilingual FAQs; instructor-approved synthetic parallel data. Students must document the source and origin of all external material.

3. Tasks & Weightage (Total: 10 Marks)

Part	Task	Marks
A	Corpus collection, preprocessing and corpus analysis	1
B	Word alignment: IBM Model 1 with Expectation–Maximization (from scratch)	2
C	Phrase-based SMT: phrase extraction, translation model, n-gram language model and decoder	3
D	Neural MT comparison using a pretrained model	2
E	Comparative evaluation and domain-specific robustness/error analysis	1
F	Reflection and deployment recommendation	1
	Total	10

4. Detailed Requirements

4.1 Corpus Preparation & Analysis (Part A)

Students must: describe the assigned business scenario; identify the target users; collect and clean the parallel corpus; perform tokenization, normalization and sentence-length filtering; split the data into training and test sets; perform a basic corpus analysis for both languages (sentence-length distribution, vocabulary size, token/type ratio).

Students must report: number of sentence pairs; vocabulary sizes; average sentence lengths; preprocessing performed; corpus limitations.

Students should justify how corpus size and domain match/mismatch affect SMT quality.

4.2 IBM Model 1 Word Alignment (Part B)

Implement IBM Model 1 from scratch (no MT toolkits may be used for this part): initialize uniform translation probabilities t(f|e); run the Expectation–Maximization algorithm for a fixed number of iterations; demonstrate convergence by plotting log-likelihood or perplexity per iteration.

Students must: display the top-5 translation candidates for at least 10 source words; visualize alignment matrices (heatmaps) for at least 3 sentence pairs; discuss at least two known limitations of Model 1 (e.g., absence of fertility, distortion/reordering modelling) with examples from their own output.

4.3 Phrase-Based SMT (Part C)

Build a phrase-based translation pipeline consisting of:

1. **Phrase extraction** — extract phrase pairs consistent with the word alignments; students may use symmetrized alignments (intersection/union heuristics).
2. **Translation model** — build a phrase translation table with translation probabilities.
3. **Language model** — train an n-gram language model (bigram/trigram with smoothing, e.g., add-k or Kneser–Ney via NLTK/KenLM) on the target side of the corpus.
4. **Decoder** — implement a simple greedy or beam-search decoder combining translation-model and language-model scores:

```
score(e) = lambda1 * log P(f|e)  +  lambda2 * log P_LM(e)
```

Students must report: number of extracted phrase pairs; language-model perplexity on held-out data; the effect of beam width on output quality and decoding time; at least 10 example translations with source, reference and system output.

Students may use NLTK utilities, but the decoder logic must be their own implementation.

4.4 Neural MT Comparison (Part D)

Translate the same held-out test set using a pretrained Neural MT model (e.g., Helsinki-NLP/opus-mt, IndicTrans2, mBART, or any instructor-approved model). Fine-tuning is NOT required.

Students must compare SMT and NMT outputs on the same sentences and consolidate the results into a table containing, where applicable:

Approach	BLEU	Adequacy (1–5)	Fluency (1–5)	OOV Handling	Latency
Phrase-Based SMT					
Pretrained NMT					

At least one comparison chart must be included.

4.5 Evaluation Dataset & Metrics (Part E)

Students must implement sentence-level and corpus-level BLEU themselves (including n-gram precision, clipping and brevity penalty), and validate their implementation against sacrebleu or NLTK.

Create an evaluation set of at least 15 sentences covering the following categories:

Sentence Category	Minimum
Short simple sentences	3
Long / syntactically complex sentences	3
Sentences with named entities and numbers	3
Sentences with domain-specific terminology	3
Out-of-domain sentences	3
Total	15

Robustness Stress Test: Each group must create at least five domain-specific tricky examples covering: negation; word-order divergence between the language pair; idioms or multi-word expressions; out-of-vocabulary words; ambiguous words with multiple senses. The examples should reflect the assigned domain's stress-test focus: headlines with dropped articles/verbs, long compound German words, named entities and quotations, date and place expressions.

SMT and NMT outputs on these examples must be compared in a table, with each error categorized (wrong lexical choice, reordering error, dropped word, hallucinated/inserted word, untranslated word).

4.6 Reflection & Deployment Recommendation (Part F)

Write a 250–400 word summary answering:

(a) Low-resource language pair, limited compute
Would you recommend SMT or NMT? Consider: data requirements; training cost; interpretability; achievable quality.

(b) The assigned high-volume production scenario
Which approach would you recommend? Consider: translation quality; cost; latency; maintainability; handling of named entities and numbers.

The recommendation must be supported by the group's experimental results. Students should also identify one major limitation of their selected approach.

5. Deliverables

One executed Jupyter notebook per GROUP. All cells must run top-to-bottom with outputs visible.

Recommended notebook naming convention:

`<PSid>_Group<GroupNo>_AIMLZG521_SMT_Assignment.ipynb`

Example:

`PS4_Group12_AIMLZG521_SMT_Assignment.ipynb`

The notebook should contain: corpus analysis; IBM Model 1 implementation with convergence plot and alignment visualizations; phrase-based SMT pipeline; NMT comparison; evaluation tables and charts; robustness analysis; reflection.

No API keys, passwords, access tokens or secrets may be committed in the submitted notebook. All external datasets, corpora, models and libraries must be appropriately cited.

6. Evaluation Rubric

Criterion	Weight	What is assessed
Correctness & completeness of Parts A–D	50%	Working corpus setup, correct IBM Model 1 EM implementation, functioning phrase-based pipeline and NMT comparison
Depth of comparative analysis	20%	Quality of experimental comparison, metrics, charts and robustness/error analysis
Reflection and deployment recommendation	15%	Depth of reasoning, course-concept application and domain-specific recommendations
Code quality & reproducibility	10%	Readable code, reusable functions, configuration, reproducibility and clear notebook narration
Research/literature component	5%	Appropriate use of research papers/technical sources and connection to the implementation
Total	100%	

Marks Conversion

The assignment is evaluated out of 10 marks.

Rubric Weight	Marks
Part A – Corpus collection, preprocessing and corpus analysis	1.0
Part B – IBM Model 1 word alignment with EM	2.0
Part C – Phrase-based SMT: translation model, language model and decoder	3.0
Part D – Neural MT comparison	2.0
Part E – Comparative evaluation and robustness analysis	1.0
Part F – Reflection and deployment recommendation	1.0

Rubric Weight	Marks
Total	10.0

7. Submission

Submit via the eLearn portal under the Assignment–2 link. ONE submission is required per group. The notebook header must contain: Problem Statement number; group number; student names; student IDs. The submitted notebook must be executable from beginning to end, with all required outputs, tables, visualizations and observations visible.

PART 2 — SITUATED LEARNING ASSIGNMENT (5 MARKS)

1. Purpose of the Assignment

This situated learning assignment is designed to assess how students recognise and use NLP applications in their working environment or the relevant domain. Students are not required to solve a predefined classroom scenario. Instead, they should investigate how a concept is currently being used or leveraged.

2. Learning Objectives

• Relate an NLP application taught in the course to an industry or organisational context.
• Identify the purpose, workflow, tools, data, and users associated with an NLP application.
• Analyse the practical benefits, limitations, and effectiveness of the application.
• Identify industry-specific challenges and opportunities for improvement.

3. Assignment Approach

CONCEPT TAUGHT → INDUSTRY USE → PRACTICAL ANALYSIS → REFLECTION

The group may select an application that is currently deployed, under development, being evaluated, or not currently used but has a realistic potential application in the organisation/industry.

4. Topics – Select ONE

1. Open-source tools used for state-of-the-art NLP applications
2. Spellcheck and Grammar Check
3. Retrieval-Augmented Generation (RAG) applications
4. Question Answering and Conversational AI
5. Sentiment Analysis
6. Privacy and Ethics in NLP Applications
7. Machine Translation
8. Information Extraction

5. Submission – Common Format

Team Details

Group Number: _____

Team Member Names: _____

BITS IDs: _____

BITS Email IDs: _____

Industry / Domain: _____

Organisation Type: _____

5.1 Industry Context – Anchor [1 Mark]

Identify how the selected NLP application is being used, being explored, or could be used in your industry or organisation. State the business/technical problem it addresses and identify the relevant users or stakeholders.

Industry / Organisation: _____

Selected NLP concept: _____

Application / Use case: _____

Problem addressed: _____

Users / stakeholders: _____

5.2 Concept in Practice – Apply [2 Marks]

Describe the practical use of the selected NLP concept. Include the data/input, NLP approach, tools/models, workflow, and output. Provide suitable evidence where possible.

Data / input: _____

NLP approach / concept: _____

Tool / model / framework: _____

Workflow / architecture: _____

Output / result: _____

Evidence (screenshots, sample output, documentation, etc.): _____

5.3 Industry Analysis – Analyse [1 Mark]

Analyse the business value of the NLP application. Identify its benefits, limitations, and effectiveness in addressing the business problem.

Key benefits / value: _____

Limitations / problems: _____

Evidence of effectiveness or impact:
_____

5.4 Reflection and Future Scope – Reflect [1 Mark]

Identify one major challenge, gap, or constraint in the current use of the concept and suggest one practical improvement or future application.

Major challenge / gap: _____

Recommended improvement: _____

Potential future use: _____

6. If the Concept Is Not Currently Used

Students should NOT invent an organisational application. If the selected concept is not currently used in the organisation, state this clearly and identify a realistic potential application based on an actual industry problem. Explain why the concept could be useful and what would be required to introduce it.

7. Evidence Expected

- Evidence of the actual industry/organisational use, where available.
- An anonymised description of the workflow, system, process, or application.
- Representative input/output, screenshots, architecture, documentation, or other suitable evidence.
- Tool/model/framework information where it can be disclosed.
- Observed benefits, limitations, or impact.
- For proposed applications: a justified industry need and realistic implementation considerations.

8. Assessment Rubric – 5 Marks

Component	Evaluation Criterion	Marks
Anchor	Relevant industry/organisational use of the selected NLP concept is clearly identified and contextualised.	1.0
Apply	Practical use is described with appropriate details of data, approach, tools/models, workflow, and evidence.	2.0
Analyse	Benefits, limitations, and effectiveness are analysed in relation to industry needs.	1.0
Reflect	A meaningful industry challenge/gap and practical improvement or future application are identified.	1.0
Total		5.0

9. Data Privacy and Academic Integrity

Students must not disclose confidential, proprietary, personally identifiable, or otherwise sensitive organisational information. Use anonymised or synthetic examples where necessary. Students should distinguish clearly between observed organisational practice and their own proposed application. Any

external tools, datasets, models, or sources used should be acknowledged.

10. Recommended Submission Length

Recommended: 2–4 pages, excluding screenshots, code, sample outputs, and supporting evidence. The focus should be on the group's understanding of how the selected NLP concept is used in an industry context.

Submit as a PDF document. Suggested file naming convention:

`Assignment2_SituatedLearning_Group_<GroupNo>.pdf`

Overall Submission & Deadline

Both components are group assignments; ONE submission per group is required for each component (notebook for Part 1, PDF for Part 2), via the eLearn portal under the Assignment–2 link. The same group composition should be maintained for both components, unless otherwise instructed. Refer to the eLearn announcement for the exact submission deadline within the assignment window specified in the course handout.

For any questions, reach out to mariyam.b@wilp.bits-pilani.ac.in.